

Hybrid Approach for Control and Fault Detection for Large Scale systems

Problem Definition

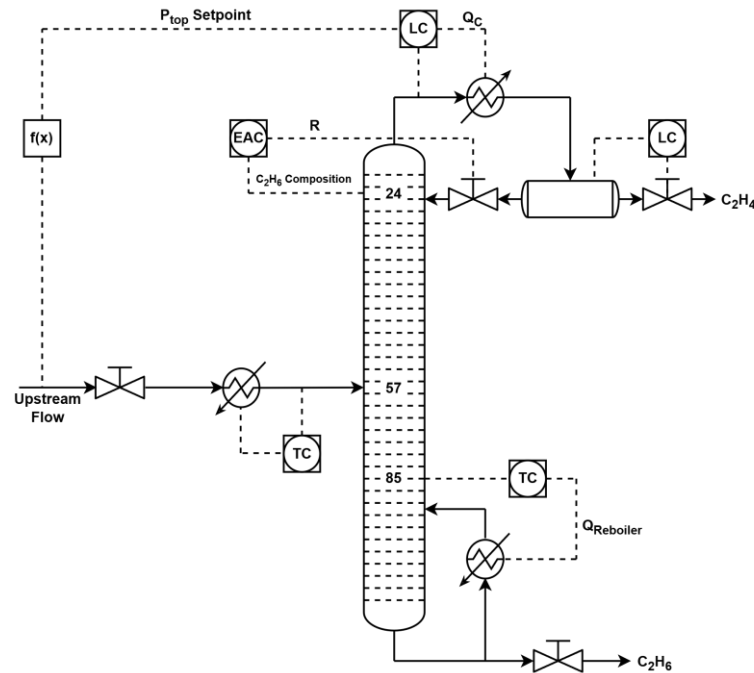
- Chemical processes must **ensure product quality, plant safety, and economic efficiency**.
- However, they are **prone to faults**, which can lead to **asset damage, unsafe operation, or plant shutdowns**.
- A robust controller is **essential** to handle disturbances and maintain system stability, **before** things go wrong.

Common Approaches & Challenges

Approach	Strengths	Limitations
PID Control	Simple, widely used in industry	Poor for MIMO/nonlinear systems
Linear MPC	Fast, convex optimization	Breaks under nonlinear dynamics, unmeasured disturbances , or faults
Nonlinear MPC	Handles nonlinearities	Computationally intensive , convergence issues
Fault Detection	Can detect after it happens	But not early enough to make proactive control decisions

Proposed Approach

- Safe RL Implementation:
 - RL learns from **MPC-generated actions** → **safe and guided** exploration.
 - Reduces the risk of unsafe behavior in industrial deployment.
- Hybrid Fault Detection, Using **first-principles hydraulic models + data-driven**:
 - Estimate **unmeasurable internal states**
 - Detect **early signs of flooding**
 - Allow **proactive control action** before faults escalate

Ethylene Splitter (C₂S)**Manipulated inputs:**

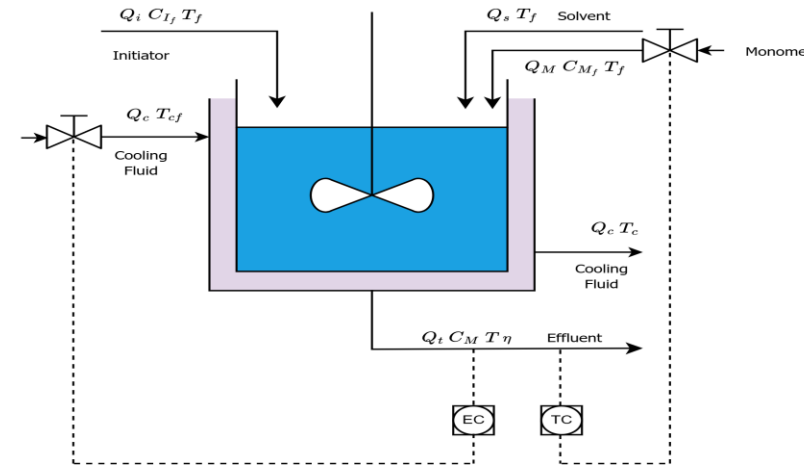
- Reflux Flow
- Reboiler Duty

Controlled Outputs:

- Tray 24 ethane comp.
- Tray 85 temp.

- Simulation is in Aspen Dynamics.
- Python and Aspen Dynamics are connected.
- Distillation Column Inputs:
 - Feed flow, pressure, and temperature
 - Feed C₂H₄ and C₂H₆ compositions

Continuous Styrene Polymerization Reactor

**Manipulated inputs:**

- Cooling flow
- Monomer Flow

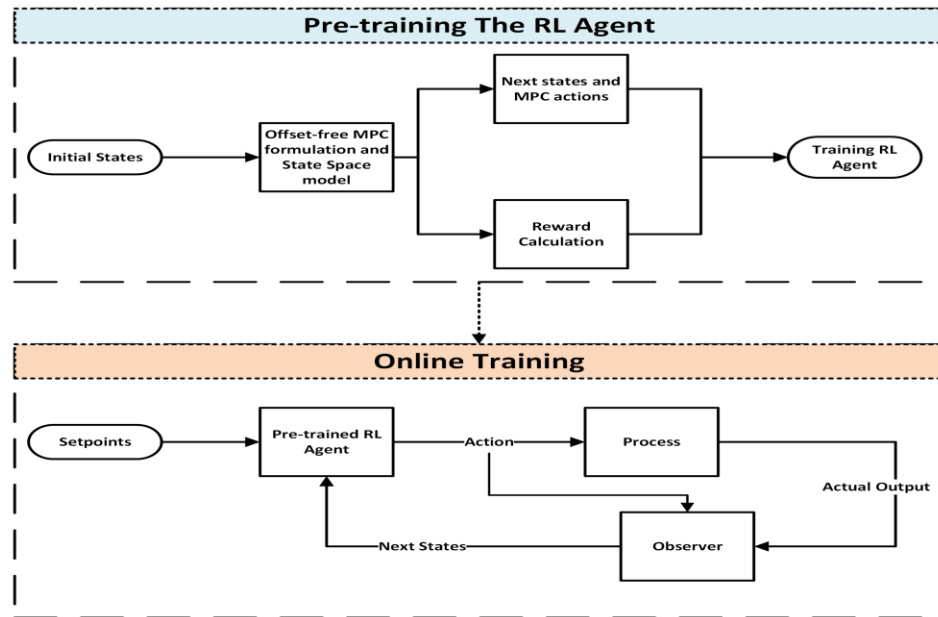
Controlled Outputs:

- Viscosity
- Temperature

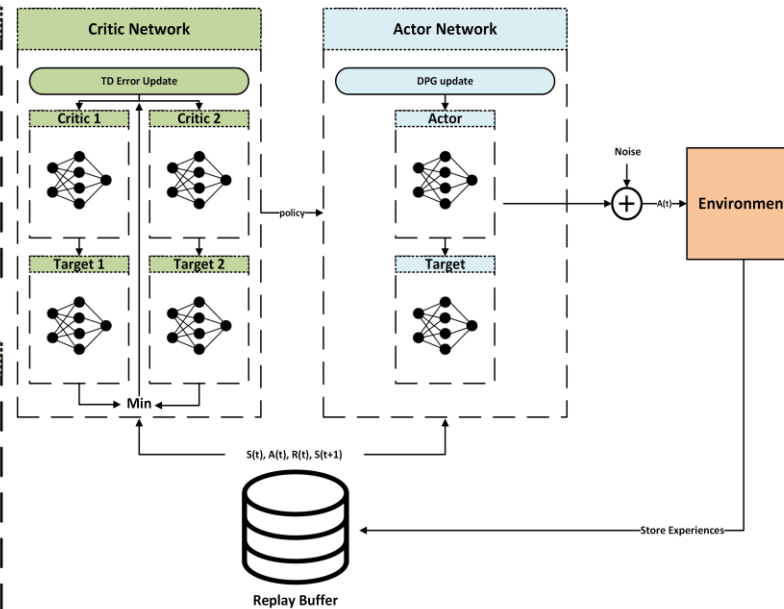
Governing Equations	Reactions
$\frac{dC_I}{dt} = \frac{Q_i C_{I,f} - Q_t C_I}{V} - k_d C_I$	$2I \rightarrow 2R \quad (k_d)$
$\frac{dT}{dt} = \frac{Q_t (T_f - T)}{V} - \frac{\Delta H}{\rho c_p} k_p C_M C_P - \frac{hA}{\rho c_p V} (T - T_c)$	$M + R \rightarrow P_1 \quad (k_i)$
$\frac{dT_c}{dt} = \frac{Q_c (T_{c,f} - T_c)}{V_c} + \frac{hA}{\rho_c c_{p,c} V_c} (T - T_c)$	$P_n + M \rightarrow P_{n+1} \quad (k_p)$
$\frac{dD_0}{dt} = \frac{1}{2} k_t C_P^2 - \frac{Q_t D_0}{V}$	$P_n + P_m \rightarrow T_n + T_m \quad (k_{td})$
$\frac{dC_M}{dt} = \frac{Q_m C_{M,f} - Q_t C_M}{V} - k_p C_M C_P$	$P_n + P_m \rightarrow T_{n+m} \quad (k_{tc})$
$\frac{dD_1}{dt} = M_m k_p C_M C_P - \frac{Q_t D_1}{V}$	Intrinsic Viscosity
$\frac{dD_2}{dt} = 5 M_m k_p C_M C_P + \frac{3 M_m k_p^2}{k_t} C_M^2 - \frac{Q_t D_2}{V}$	

$$k = A \cdot \exp\left(-\frac{E}{T}\right), Q_t = Q_i + Q_s + Q_m, C_P = \sqrt{\frac{2 f_i k_d C_I}{k_t}}$$

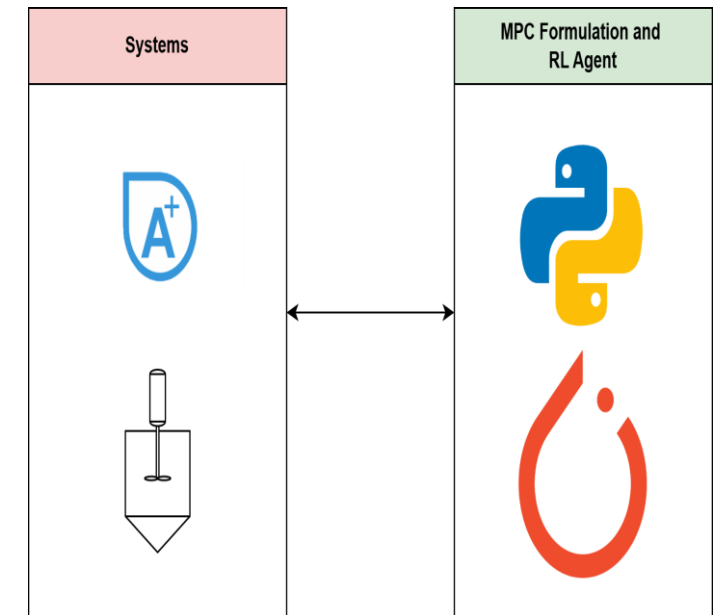
Safe RL Control Implementation



Block Diagram of RL Controller Design



TD3 Schematic



Working Environment

Pre-train Phase

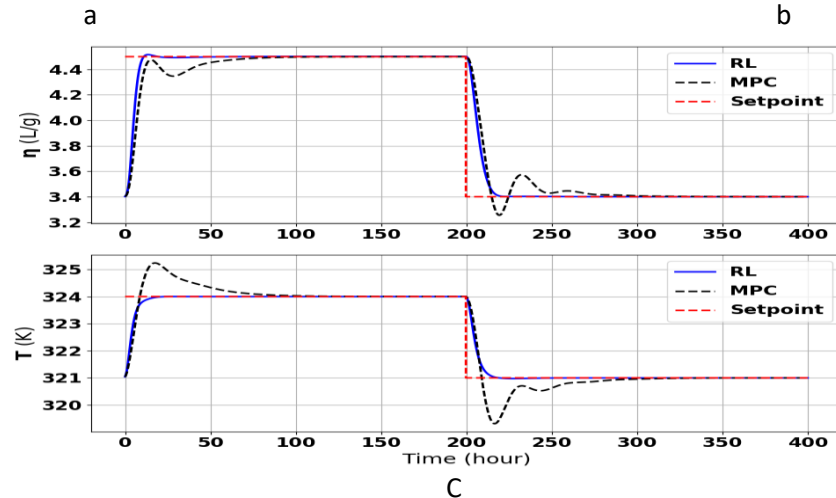
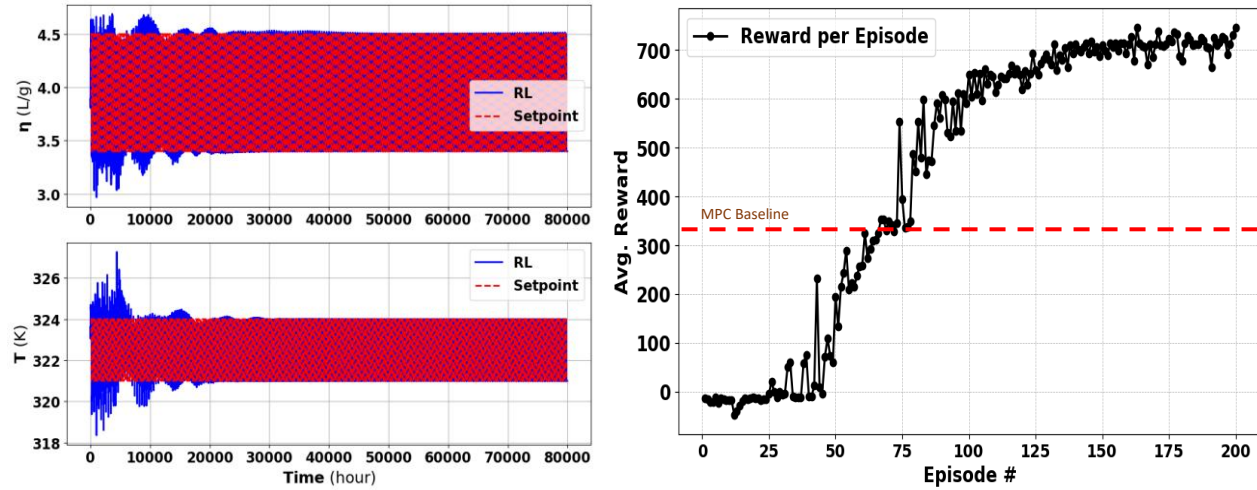
1. Linear state-space model from step test
2. Create random states in range of max and min of the states
3. Solve MPC formulation and get the control action
4. Next state and output from the linear state space model
5. Calculate reward
6. Stack all these experiences in the replay buffer
7. Pre-train TD3 Agent

Online deployment Phase

1. Adjustments to TD3 Agents
 1. An exponential bonus reward function to encourage offset free behavior
 2. Huber loss function to learn more from undesired actions
2. Warm-start: Let the agent with restricted exploration to search over the action boundaries
3. After warm-start the agent will start training to find the optimal policy
4. Exploration noise will decay over time
5. The agent will train under nominal conditions and disturbances

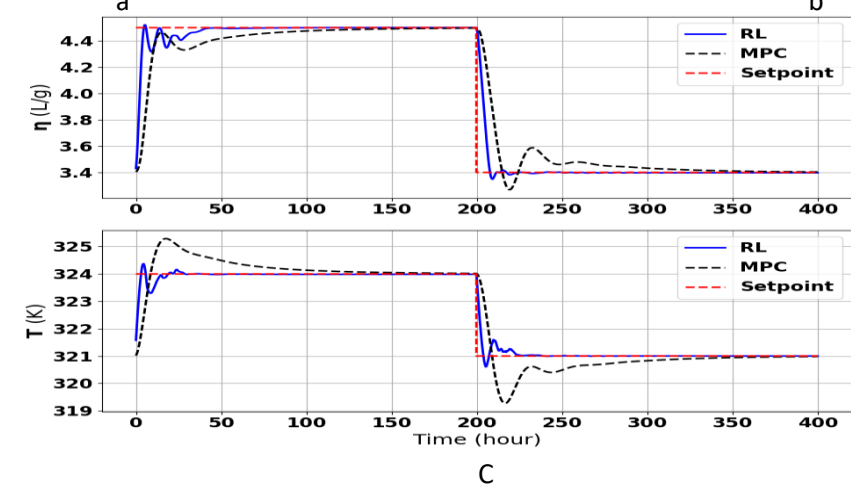
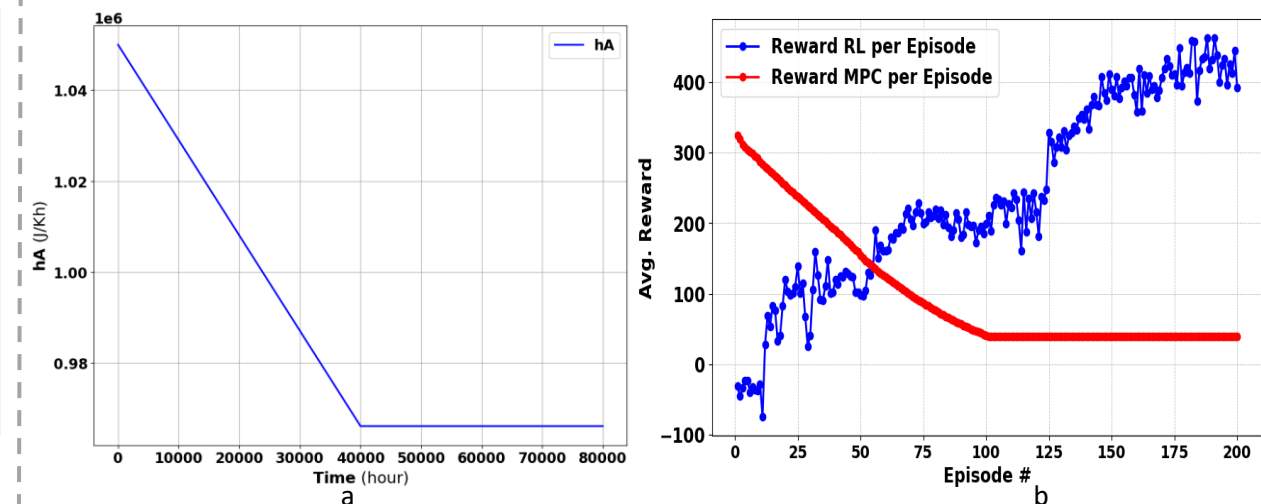
Proposed RL agent controller Results for Polymer Reactor

Nominal Conditions



a) RL setpoint tracking over 200 episodes b) Average rewards vs. time c) RL vs. MPC for last episode

Under Disturbances



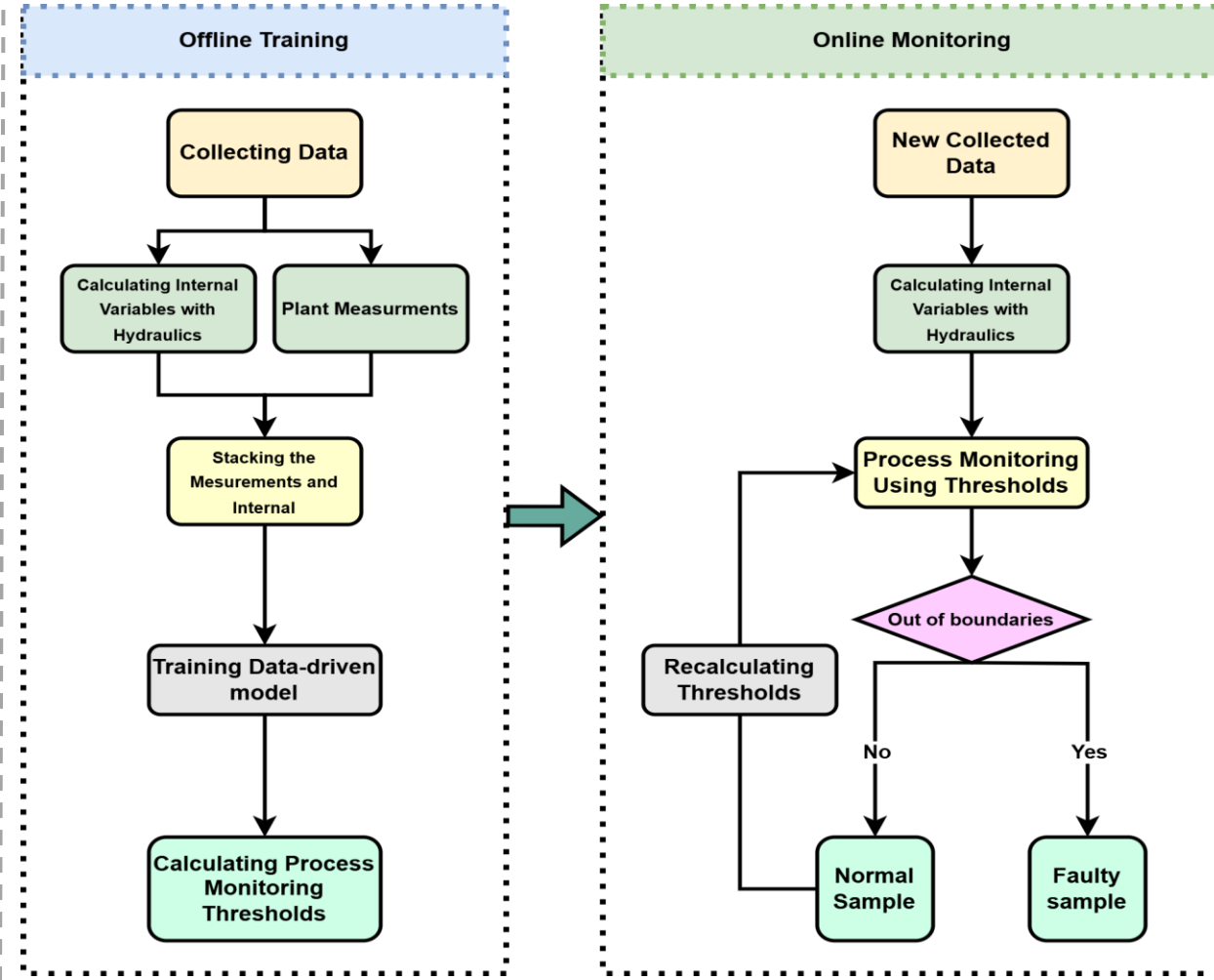
a) Disturbance b) Average rewards vs. time c) RL vs. MPC for last episode

Fault Detection

ODEs Distillation column Hydraulic for first-principle

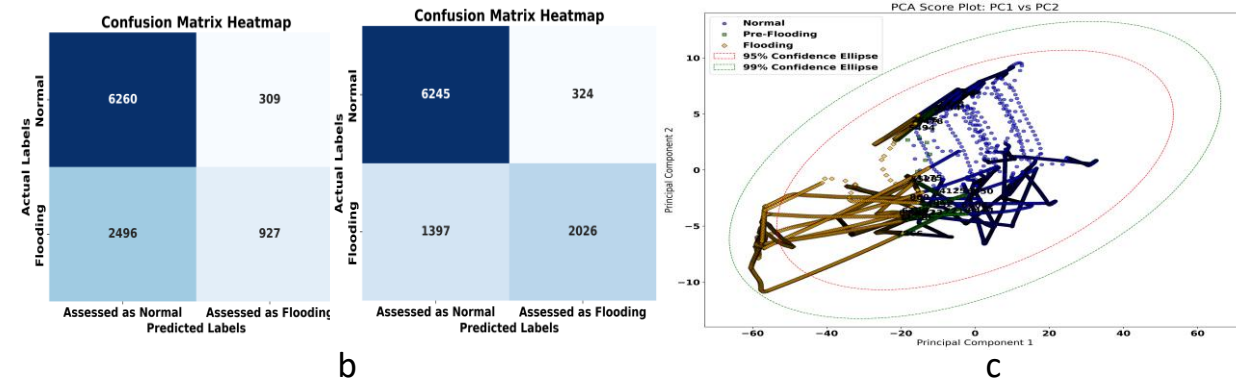
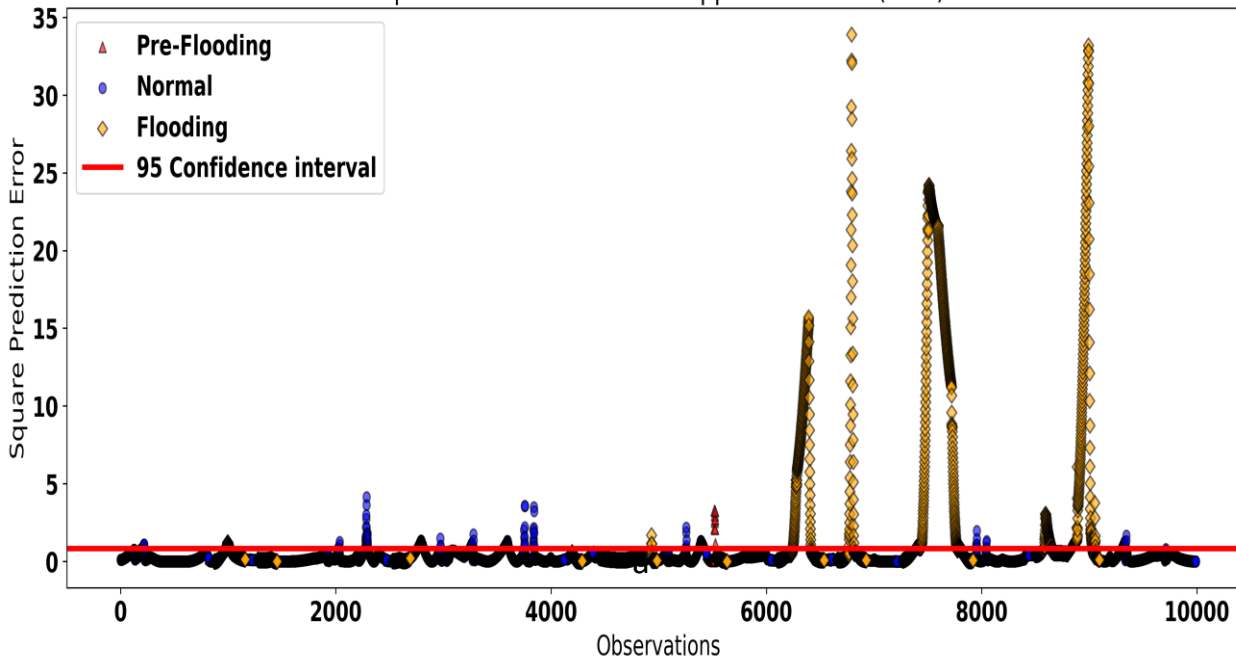
Francis Weir Flow Law	$L_i = \begin{cases} 0, & \text{if } h_{f,i} \leq h_{\text{weir}} \\ \frac{2}{3} C_{\text{weir}} \ell_{\text{weir}} \sqrt{2g} (h_{f,i} - h_{\text{weir}})^{1.5}, & \text{if } h_{f,i} > h_{\text{weir}} \end{cases}$
Froth Level (Aerated)	$\epsilon_i = \frac{A_f(u_{v,i})^{B_f}}{1 + A_f(u_{v,i})^{B_f}} \text{ and } h_{f,i} = \frac{h_i}{1 - \epsilon_i}$
Vapor Flow	$V_i = \begin{cases} V_{\text{above}}, & \text{if } i \leq i_F \\ V_{\text{below}}, & \text{if } i > i_F \end{cases} \text{ and } u_{v,i} = \frac{V_i}{A_{\text{tray}}}$
Down-comer out flow	$D_i = \begin{cases} C_d A_o \sqrt{2g \Delta h_i}, & \text{if } \Delta h_i > 0 \\ 0, & \text{if } \Delta h_i \leq 0 \end{cases} \text{ where } \Delta h_i = h_{\text{dc},i} - h_{f,i+1}$
Effective Liquid Flow	$\tilde{L}_i = \begin{cases} L_i, & \text{if } V_{d,i} < V_D^{\text{max}} \\ \min(L_i, D_i), & \text{if } V_{d,i} \geq V_D^{\text{max}} \end{cases}$
System of ODEs	$h_i(t) \text{ for } i = 1, \dots, n \text{ and } V_{d,i}(t) \text{ for } i = 1, \dots, n-1$
Top Tray Liquid Height	$\frac{dh_1}{dt} = \frac{1}{A_T} (R - \tilde{L}_1)$
Middle Tray Liquid Height	$\frac{dh_i}{dt} = \frac{1}{A_T} (D_{i-1} + \delta_{i,i_F} F - \tilde{L}_i)$
Bottom Tray Liquid Height	$\frac{dh_n}{dt} = \frac{1}{A_T} (D_{n-1} - \tilde{L}_n)$
Down-comer Filling	$\frac{dV_{d,i}}{dt} = \tilde{L}_i - D_i, \quad i = 1, \dots, n-1$

Proposed Approach



Results

Square Prediction errors vs Upper Threshold (95%)



a) SPE Chart for Process Monitoring b) Confusion Matrix c) Score Plot (PC1 vs. PC2)

Ongoing and Future Works

Ongoing:

- RL implementation on the distillation column
- Applying soft and hard constraints on the RL controller
- Test proposed approach on more disturbances
- Adding flash calculation to the hydraulics
- Improving predicting capability of the fault detection

Future Works:

- Combine Fault detection and RL controller to have an agent that prevent faults
- RL-Assisted MPC which improve MPC performance by adjusting MPC and model parameters during the process
- RL behavioral cloning and compare to pre-training